6. Evaluate $\int_{-\pi/2}^{\pi/2}\int_{0}^{2\cos\theta} r^2 dr d\theta$.

7. Form the differential equation by eliminating the orbitary constant from $y^2 = 4ax$.

8. Solve $ydx + xdy = e^{-xy}dx$ if cuts the y-axis.

9. Solve $(D^2 - 6D + 9)y = e^{3x}$.

10. Solve $(x^2D^2 + 3xD + 1)y = 0$.

PART B — (5 × 11 = 55 marks)

Answer ONE full question from each unit.

UNIT I

11. (a) Find the radius of curvature at any point $(a\cos^3\theta, a\sin^3\theta)$ on the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$. (6)

(b) Find the equation of the circle of curvature at (c, c) on $xy = c^2$. (5)

Or

12. Show that evaluate of the curve $x = a(\cos\theta + \theta\sin\theta)$, $y = a(\sin\theta - \theta\cos\theta)$ is a circle. (11)

UNIT II

13. (a) If $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ evaluate $\frac{\partial(r, \theta)}{\partial(x, y)}$. (5)

(b) Expand $x^2y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$ using Taylor's expansion. (6)

Or

14. Find the maxima and minima of $f(x, y) = \sin x \sin y \sin(x + y)$, $0 \le x, y < \pi$. (11)

UNIT III

15. Change the order of integration and hence evaluate $\int_{0}^{1}\int_{x^2}^{2-x} xy\, dy\, dx$. (11)

Or

16. By changing into polar co-ordinates show that $\int_{0}^{\infty}\int_{0}^{\infty} e^{-(x^2+y^2)}dx\, dy = \frac{\pi}{4}$. Hence evaluate $\int_{0}^{\infty} e^{-t^2dt} = \frac{\sqrt{\pi}}{2}$.

(11)

UNIT IV

17. (a) Solve the differential equation $\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^{-1}y}{1+y^2}$. (5)

(b) Solve $\frac{dy}{dx}=\sin(x+y)$. (6)

Or

18. (a) Solve the differential equation $\cos^2 x dy + y e^{\tan x} dx = 0$. (5)

(b) Find the equation of curve passing through $(1, 0)$ and which has slope $1+\frac{y}{x}$ at (x, y).

UNIT V

19. (a) Solve $(x^2D^2 + xD)y = \log x$. (6)

(b) Solve $(D^2 - 2D - 8)y = \cos 2x$. (5)

Or

20. Solve the simultaneous equations $\frac{dx}{dt}+y=\sin t+1$ and $\frac{dy}{dt}+x=\cos t$ given that $x(0)=1$ and $y(0)=2$. (11)

5781001

B.Tech. DEGREE EXAMINATION,
AUGUST/SEPTEMBER 2022.

First Semester

Common to All Branches

MATHEMATICS – I

(From 2013 – 14 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Find the radius of curvature at any point (x, y) on the curves $y = c\log\sec\left(\frac{x}{c}\right)$.

2. Define Beta function and Gamma function.

3. Find $\frac{d^2y}{dx^2}$ for $x = a\cos\theta$; $y = b\sin\theta$.

4. Expand x^y near the point $(1, 1)$ upto 1st degree using Taylor's expansion.

5. Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dy\, dx}{\sqrt{a^2-x^2-y^2}}$.